

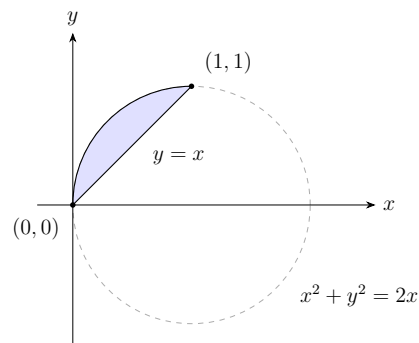
1. **Final answer:** $\boxed{\int_{\pi/4}^{\pi/2} \int_0^{2 \cos \theta} r^3 dr d\theta}.$

The Cartesian bounds and the corresponding polar bounds are

$$\begin{cases} 0 \leq x \leq 1, \\ x \leq y \leq \sqrt{2x - x^2} \end{cases} \implies \begin{cases} \pi/4 \leq \theta \leq \pi/2, \\ 0 \leq r \leq 2 \cos \theta. \end{cases}$$

The circle is $(x - 1)^2 + y^2 = 1$, or $r = 2 \cos \theta$. The retained part lies above $y = x$, so its angles run from $\pi/4$ to $\pi/2$. The boundaries meet at $(0, 0)$ and $(1, 1)$. These polar bounds also give $x = r \cos \theta \leq 2 \cos^2 \theta \leq 1$.

Since $x^2 + y^2 = r^2$ and $dA = r dr d\theta$, the transformed integrand is r^3 . No evaluation is requested.



2. **Final answer:** $\boxed{M = 4\pi/3}.$

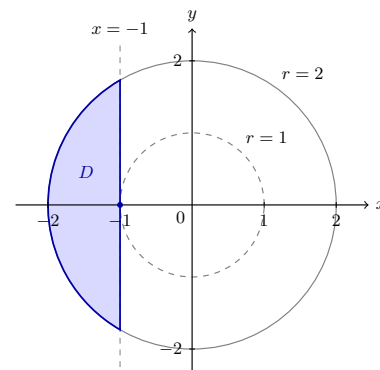
For $-2 \leq x \leq -1$, the condition $x^2 + y^2 \geq 1$ is automatic. The inner circle touches the retained region only at $(-1, 0)$. Thus the Cartesian and polar bounds are

$$\begin{cases} -2 \leq x \leq -1, \\ -\sqrt{4 - x^2} \leq y \leq \sqrt{4 - x^2} \end{cases} \implies \begin{cases} 2\pi/3 \leq \theta \leq 4\pi/3, \\ -\sec \theta \leq r \leq 2. \end{cases}$$

The line $x = -1$ becomes $r \cos \theta = -1$. Since $\cos \theta < 0$ here, dividing $r \cos \theta \leq -1$ reverses the inequality: $r \geq -\sec \theta$. At $r = 2$ the line gives $\cos \theta = -1/2$, determining the angular endpoints. Also, $-\sec \theta \geq 1$ throughout this interval.

Multiplying the density $r^2 - 1$ by the polar area factor r gives

$$M = \int_{2\pi/3}^{4\pi/3} \int_{-\sec \theta}^2 (r^2 - 1)r dr d\theta.$$



Using $\int \sec^4 \theta d\theta = \tan \theta + \frac{1}{3} \tan^3 \theta + C$,

$$\begin{aligned} M &= \int_{2\pi/3}^{4\pi/3} \left(2 - \frac{1}{4} \sec^4 \theta + \frac{1}{2} \sec^2 \theta \right) d\theta \\ &= \left[2\theta + \frac{1}{4} \tan \theta - \frac{1}{12} \tan^3 \theta \right]_{2\pi/3}^{4\pi/3} = \frac{4\pi}{3}. \end{aligned}$$

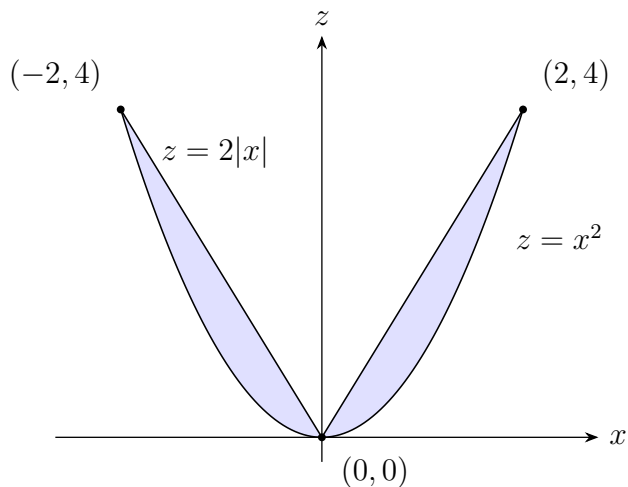
At either endpoint, $\tan \theta = \pm\sqrt{3}$, so $\frac{1}{4} \tan \theta - \frac{1}{12} \tan^3 \theta = 0$.

3. **Final answer:** the two shaded lobes below, with intersections $(x, z) = (-2, 4), (0, 0), (2, 4)$.

The given cylindrical bounds and their restriction to $y = 0$ are

$$\begin{cases} 0 \leq \theta \leq 2\pi, \\ 0 \leq r \leq 2, \\ r^2 \leq z \leq 2r \end{cases} \implies \begin{cases} -2 \leq x \leq 2, \\ x^2 \leq z \leq 2|x|. \end{cases}$$

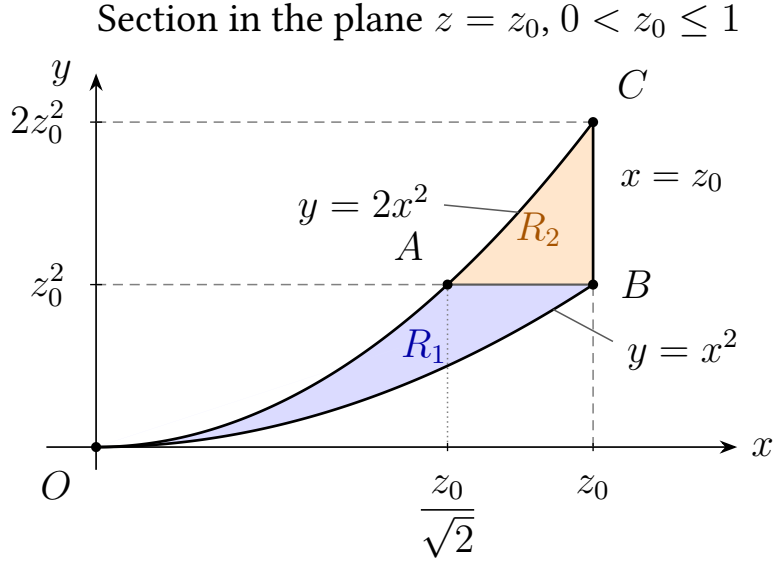
In Cartesian coordinates the solid satisfies $x^2 + y^2 \leq 4$ and $x^2 + y^2 \leq z \leq 2\sqrt{x^2 + y^2}$. On the entire xz -plane, $r = \sqrt{x^2} = |x|$. Consequently, the section has lower boundary $z = x^2$ and upper boundary $z = 2|x|$. Solving $x^2 = 2|x|$ gives $x = -2, 0, 2$.



4. **Final answer:**

$$I = \int_0^1 \int_0^z \int_{x^2}^{2x^2} z^2 x^3 e^{x^3} dy dx dz = \frac{3-e}{9}.$$

Fix a height $z = z_0$, with $0 < z_0 \leq 1$, and view that horizontal plane in (x, y) -coordinates. The first integral covers R_1 (blue), and the second covers R_2 (orange):



The pieces share the segment AB on $y = z_0^2$; BC is the boundary $x = z_0$. The line $y = 2z_0^2$ meets the region only at C . The labeled coordinates are (x, y) ; append z_0 to obtain points in the solid. At $z_0 = 0$, the section collapses to the origin.

For each fixed $0 \leq z \leq 1$, the two original sections combine as follows:

$$\begin{cases} 0 \leq y \leq z^2, \\ \sqrt{y/2} \leq x \leq \sqrt{y} \end{cases} \quad \text{or} \quad \begin{cases} z^2 \leq y \leq 2z^2, \\ \sqrt{y/2} \leq x \leq z \end{cases} \iff \begin{cases} 0 \leq x \leq z, \\ x^2 \leq y \leq 2x^2. \end{cases}$$

The curves are $y = x^2$ and $y = 2x^2$; the right boundary is $x = z$. The original upper x -bound changes from $\sqrt{y}$ to z at $y = z^2$. The two pieces meet only along that boundary, so their integrals add without double-counting volume. No coordinate change or Jacobian is needed.

Integrating in y supplies $2x^2 - x^2 = x^2$. Next use $u = x^3$ and integration by parts, with $\int u e^u du = (u - 1)e^u + C$:

$$\begin{aligned} I &= \int_0^1 z^2 \int_0^z x^5 e^{x^3} dx dz = \frac{1}{3} \int_0^1 z^2 ((z^3 - 1)e^{z^3} + 1) dz \\ &= \frac{1}{9} \int_0^1 ((v - 1)e^v + 1) dv = \frac{1}{9} [(v - 2)e^v + v]_0^1 = \frac{3 - e}{9}, \quad v = z^3. \end{aligned}$$